Every fixed tree has a properly colored spanning subdivision

Ruonan Li, Ruhui Lu, Xueli Su, Shenggui Zhang

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Editorial note. Original complete author body retained to preserve definitions, supporting arguments and figure references; exactly one primary is counted per task. No mathematical repair, added hypothesis or correctness certification.

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Allowed standard prerequisites (editorial).

- Graph coloring and directed graph terminology. Native original definitions, auxiliary results and cited standard references retained in the accompanying author text.. The counted embedding or extension argument is retained in full.

1 Introduction

Let G be an edge-colored graph. We say G is *monochromatic* if all the edges are of the same color, and *rainbow* if all the edges are of distinct colors, and *lexical* if there is a total order of $V(G)$ such that edges uv and xy have a same color if and only if $\min\{u, v\} = \min\{x, y\}$. The *Canonical Ramsey Theorem* founded by Erdős and Rado [12] states that given an integer k, each edge-colored complete graph of sufficiently large order always contains a k-clique H, which is either monochromatic or rainbow or lexical. In fact, the absence of certain rainbow (monochromatic) subgraph often forces a giant connected subgraph that is monochromatic (colors fully mixed). For instance, Erdős and Rado [12] observed that every 2-colored complete graph always contains a monochromatic spanning tree. Gallai partition theorem [14](c.f.[17]) implies that the absence of rainbow triangle in edge-colored complete graphs forces a spanning 2-colored subgraph. Studying a conjecture proposed by Andersen [4], Alon, Pokrovskiy and Sudakov [3] proved that each edge-colored K_n without monochromatic path of length 2 contains a rainbow path of length $n - o(n)$. Asymptotically solving a conjecture given by Bollobás and Erdős[8], Lo's result [25] tells that each edge-colored K_n without monochromatic star of size $(1/2 - \epsilon)n$ always contains a properly colored Hamilton cycle when n is sufficiently large. An edge-colored graph is called *properly colored* (say "PC " for short) if every two adjacent edges are of distinct colors. In this paper, we continue the exploration of spanning properly colored subgraphs in edge-colored complete graphs when a fixed monochromatic configuration is forbidden.

An easy observation given by Barr [5] tells that each edge-colored K_n without containing a monochromatic triangle (say "mono-C_3-free" for short) must contain a PC Hamilton path. Under the same condition, the first author [22] studied the existence of PC Hamilton cycles and obtained a full characterization of the counterexamples. Particularly, when $n \geq 6$, every counterexample acts locally like a non-strongly-connected directed graph. In fact, for any PC spanning target graphs containing cycles, an edge-colored complete graph transformed from a transitive tournament is always a counterexample. Therefore it is natural to study the existence of properly colored spanning trees in

mono-C_3-free complete graphs. For more relations between edge-colored graphs and directed graphs, we refer the readers to [1, 11, 13, 23, 24].

Let T_0 be a fixed tree. We use $\mathcal{T}(n, T_0)$ to denote the collection of n-vertex trees that are subdivisions of T_0. Note that a Hamilton path can be regarded as a subdivision of K_2. We show the existence of a properly colored spanning tree which is a subdivision of a given tree.

Theorem 1. *Let T_0 be a tree of k edges and let G be a mono-C_3-free edge-colored K_n with $n \geq (k+2)!$. Then there exists a tree $T \in \mathcal{T}(n, T_0)$ such that G contains a PC copy of T.*

In the above theorem, there is no control on the distribution of subdividing vertices on different edges of T_0. We wonder the existence of all possible subdivisions and therefore propose the following conjecture.

Conjecture 1. Let T_0 be a fixed tree. Then there is a function $f(T_0)$ such that every mono-C_3-free edge-colored K_n with $n \geq f(T_0)$ contains a PC copy of T for each tree $T \in \mathcal{T}(n, T_0)$.

We confirm the above conjecture when T_0 is a star. For $k \geq 3$, a subdivision of a k-star is called a *k-spider*. A *leg* in a k-spider T is a path P from the unique k-degree vertex to a leaf of T. If P contains ℓ edges, then we say the leg P is of length ℓ. Use C_3 and S_k to denote a triangle and a star of k edges. Let $g(S_k, C_3)$ to be the maximum number N such that there exists an edge-colored K_N containing neither a rainbow S_k nor a monochromatic C_3. The existence of $g(S_k, C_3)$ is guaranteed by the Canonical Ramsey Theorem. In literature, Gyárfás, Lehel, Schelp and Tuza [16] studied the *Local Ramsey number*; Jamison, Jiang and Ling [20] defined the *Constraint Ramsey Number*. Both definitions are consistent with $g(S_k, C_3)$. We will use the notation $g(S_k, C_3)$ to state our main result. A weak upper bound of $g(S_k, C_3)$ is given in Section 4, which is applied in the proof of Theorem 1.

Theorem 2. *Given an integer $k \geq 3$, let G be a mono-C_3-free edge-colored K_n. If $n \geq 6k \cdot g(S_k, C_3) + 2k^3 + 2k^2 + 8k$, then for every k positive integers $\ell_1 \geq \ell_2 \geq \cdots \geq \ell_k$ satisfying $\sum_{i=1}^{k} \ell_i = n - 1$, G contains a properly colored spanning spider T with legs of lengths $\ell_1, \ell_2, \ldots, \ell_k$, respectively.*

Sketch of the proofs. To prove Theorem 1, we first show the existence of every properly colored tree on k edges (this is guaranteed by the Canonical Ramsey Theorem for sufficiently large n. Our proof gives an explicit bound for n), and then embed the remaining vertices greedily. However, the greedy method can not guarantee the number of subdividing vertices on each edge precisely as whatever we want. Therefore the key point in the proof of Theorem 2 is using some extensible structures as glue to merge vertices in mono-C_3-free complete graphs into legs. If the host graph has many "nice bowties", then we are home. Otherwise, by removing constant number of vertices, the host graph is essentially a multipartite tournament with certain properties inherent from the "mono-C_3-free" condition. Then by analyzing the structure of this multipartite tournament, we obtain a certain oriented tree which is almost the spanning tree we desired, except for the first leg. Applying some structural lemmas proved in Section 2, we finally embed the remaining vertices.

In Section 2, we define some crucial structures and obtain related properties. In Section 3, we study the "mono-C_3-free tournament" and show the existence of a certain oriented tree which is almost spanning. The proofs of Theorems 1 and 2 are deliverd in Section 4.

2 Preliminaries

2.1 Related notions

Let G be an undirected graph. For each $U \subseteq V(G)$, we write $G - U$ for $G[V(G) \setminus U]$. If $U = \{v\}$ is a singleton, we write $G - v$ rather than $G - \{v\}$. Instead of $G - V(G')$ we simply write $G - G'$. For a set of edges F, we define $G - F := (V(G), E(G) \setminus F)$ and $G + F := (V(G) \cup V(F), E(G) \cup F)$; as above, $G - \{e\}$ and $G + \{e\}$ are abbreviated to $G - e$ and $G + e$. Let P be a path in G. The length of P is the number of edges on P.

Let G be an edge-colored graph. Denote by $col(e)$ and $col(G)$, respectively, the color of an edge e and the set of colors assigned to $E(G)$. For a vertex $v \in V(G)$, the *color degree* of v in G, denoted by

$d^c_G(v)$ is the number of distinct colors assigned to the edges incident to v. We use $\delta^c(G) = \min\{d^c_G(v) : v \in V(G)\}$ to denote the *minimum color degree* of G, and $\Delta^{mon}_G(v) = \max_{c\in col(G)} |\{u \in V(G) \setminus \{v\} : col(uv) = c\}|$ to denote the *maximum monochromatic degree* of v. For two disjoint subsets V_1 and V_2 of $V(G)$, denote by $col(V_1, V_2)$ the set of colors appearing on the edges between V_1 and V_2 in G. When $V_1 = \{v\}$, use $col(v, V_2)$ to denote $col(\{v\}, V_2)$.

Let D be a directed graph. We use $V(D)$ and $A(D)$ to denote the vertex set and the arc set of D. If $uv \in A(D)$, then we say u *dominates* v, and denote it by $u \to v$. For two disjoint subsets X, Y of $V(D)$, if each arc uv between X and Y satisfies that $u \in X$ and $v \in Y$, then we say $X \to Y$. For a vertex $u \in V(D)$, denote by $N^+_D(u)$ the set of vertices that are dominated by u, and denote by $N^-_D(u)$ the set of vertices that are dominating u. The cardinality of $N^+_D(u)$ and $N^-_D(u)$ are denoted by $d^+_D(u)$ and $d^-_D(u)$, respectively. When this is no ambiguity, we often omit the subscript D. Let $P = v_1v_2\cdots v_{t+1}$ be a directed path in D with $v_i \to v_{i+1}$ for $i = 1, 2, \ldots, t$. We say the length of P is t and the directed path P is starting at v_1 and ending at v_{t+1}.

For other notations and terminologies not defined here, we refer the reader to [9].

2.2 Extensible structures

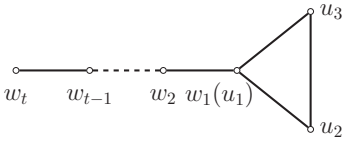
The main challenge for embedding PC spanning trees is to merge pieces of PC structures together. We find the following configurations can be used as glue for extending a PC path with certain constraints.

Definition 1 (**Center edge of a colored triangle**). Let S be an edge-colored triangle, if $col(e) \notin col(S - e)$ for an edge e in S, then we say e is a *center edge* of S.

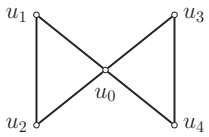
Apparently, every rainbow triangle contains 3 center edges, every 2-colored triangle contains a unique center edge and every monochromatic triangle contains no center edge.

Definition 2 (**Nice t-shovel**). Let Y be an edge-colored graph, consisting of a path $P = w_1w_2\cdots w_t$ ($t \geq 1$) and a triangle $S = u_1u_2u_3u_1$ with $V(P) \cap V(S) = \{w_1\} = \{u_1\}$, as shown in the Figure 1(a). We call Y a *t-shovel*. If P is a PC path, u_2u_3 is a center edge of S and $col(w_1w_2) \notin col\,(w_1, \{u_2, u_3\})$, then we say Y is a *nice t-shovel*. For a nice t-shovel Y with $t \geq 2$, we call $u_1 = w_1$ the *center* of Y.

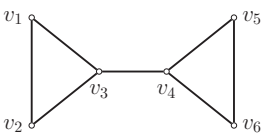
Definition 3 (**Nice bowtie**). Let F_1 be a graph obtained by identifying one vertex of two disjoint triangles (see Figure 1(b)) and let F_2 be a graph obtained by adding an edge between two disjoint triangles (see Figure 1(c)). Then we call both F_1 and F_2 *bowtie*. To be specific, we say F_1 is a *short bowtie* and F_2 is a *long bowtie*. Let F_1 and F_2 be labeled as Figure 1(b) and Figure 1(c). We say u_0 is the *center* of F_1 and u_3, u_4 are centers of F_2. If u_1u_2 and u_3u_4 in F_1 are center edges of their lying triangles and $col(u_0, \{u_1, u_2\}) \cap col(u_0, \{u_3, u_4\}) = \emptyset$, then we say F_1 is a *nice short bowtie* (or nice bowtie). If $F_2 - \{v_1, v_2\}$ and $F_2 - \{v_5, v_6\}$ are nice shovels, then we say F_2 is a *nice long bowtie* (or nice bowtie).



(a) A t-shovel



(b) A short bowtie F_1 with u_0 as a center



(c) A long bowtie F_2 with v_3 and v_4 as centers

Figure 1: Extensible structures

The following lemmas are useful in the proofs of the main theorems.

Lemma 1. *Let G be a mono-C_3-free edge-colored K_n and let $P = v_1v_2\cdots v_t$ be a PC path in G with $t \geq 2$. If a vertex $v \in V(G) \setminus V(P)$ satisfies $col(vv_i) = col(v_iv_{i+1})$ for some $i \in [1, t-1]$, then there exists $j \in [i, t-1]$ such that $Q = v_1v_2\cdots v_jvv_{j+1}\cdots v_t$ is a PC path with $col(vv_j) = col(v_jv_{j+1})$.*

Proof. Let j be the largest integer in $[1, t-1]$ satisfying $col(vv_j) = col(v_jv_{j+1})$. Then $j \geq i$. Since $vv_jv_{j+1}v$ is not a monochromatic C_3, we have $col(vv_j) \neq col(vv_{j+1})$. If $j = t-1$, then $Q = v_1v_2 \cdots v_{t-1}vv_t$ is a desired PC path. If $j \leq t-2$, then by the maximality of j, we have $col(vv_{j+1}) \neq col(v_{j+1}v_{j+2})$, which implies $Q = v_1v_2 \cdots v_jvv_{j+1} \cdots v_t$ is a desired PC path. □

Lemma 2. *Let G be a mono-C_3-free edge-colored K_n. Then for each vertex $v \in V(G)$, there is a PC Hamilton path in G starting at v.*

Proof. Fix a vertex v, let $P = v_1v_2 \cdots v_t$ be a longest PC path in G with $v_t = v$. Suppose, to the contrary, that P is not a Hamilton path. Then for a vertex $u \in V(G) \setminus V(P)$, there holds $col(uv_1) = col(v_1v_2)$. Apply Lemma 1 to the pair (P, u). We get a longer PC path starting at $v_t = v$, a contradiction. □

Lemma 3 (Spanning nice shovel)**.** *Let G be a mono-C_3-free edge-colored K_n with $n \geq 3$. Then G contains a spanning subgraph which is a nice $(n-2)$-shovel.*

Proof. When $n = 3$, it is trivial. When $n = 4$, if G has a vertex v satisfying $d^c(v) = 2$ or G is a PC graph, then we can obtain a nice shovel immediately. Otherwise since no monochromatic triangle exists, G has only one vertex of color degree 1 and three vertices of color degree 3, which also implies the existence of a spanning nice shovel.

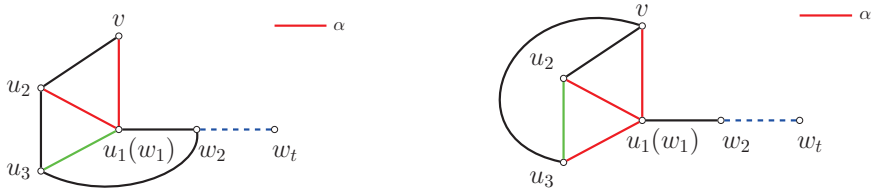
Now consider the case $n \geq 5$. Let Y be a largest PC shovel contained in G with $S = u_1u_2u_3u_1$ being the triangle in Y, $u_1 = w_1$ and $P = w_1w_2 \cdots w_t$ being the remaining path in Y. The above analysis implies $t \geq 2$. Suppose that $V(Y) \neq V(G)$. Let v be a vertex in $V(G) \setminus V(Y)$.

Case 1. $col(vu_1) \notin col(u_1, \{u_2, u_3\})$.

For each $j \in [1, t-1]$, define $P_j = w_1w_2 \cdots w_jvw_{j+1} \cdots w_t$ and define $P_t = w_1w_2 \cdots w_tv$. If P_j is a PC path for some $j \in [2, t]$, then P_j and the triangle $u_1u_2u_3u_1$ form a nice shovel larger than Y, a contradiction. So none of the paths $P_t, P_{t-1}, \ldots, P_2$ is a PC path. This implies that $col(vw_j) = col(w_jw_{j-1})$ for all $j \in [2, t]$. Since the triangle vw_2w_1v is not monochromatic we have $col(vw_2) \neq col(vw_1)$. Therefore $P_1 = w_1vw_2 \cdots w_t$ and the triangle $u_1u_2u_3u_1$ form a nice shovel larger than Y, a contradiction.

Case 2. $col(vu_1) \in col(u_1, \{u_2, u_3\})$.

Without loss of generality, assume that $col(vu_1) = col(u_1u_2) = \alpha$. For each $j \in [2, t-1]$, define $P_j = w_1w_2 \cdots w_ju_3w_{j+1} \cdots w_t$ and define $P_t = w_1w_2 \cdots w_tu_3$. If P_j is a PC path for some $j \in [2, t]$, then P_j and the triangle $u_1u_2vu_1$ form a nice shovel larger than Y, a contradiction. So none of the paths $P_t, P_{t-1}, \ldots, P_2$ is a PC path. This implies that $col(u_3w_j) = col(w_jw_{j-1})$ for all $j \in [2, t]$. Since the triangle $u_3w_2w_1u_3$ is not monochromatic we have $col(u_3w_2) \neq col(u_3w_1)$. If $col(u_3w_1) \neq \alpha$, then $P_1 = w_1u_3w_2 \cdots w_t$ and the triangle $u_1u_2vu_1$ form a nice shovel larger than Y(see Figure 2(a)), a contradiction. Therefore $col(u_3w_1) = col(u_2w_1) = col(vw_1) = \alpha$ and $\alpha \neq col(w_1w_2)$. Now consider the triangle vu_2u_3v. None of the three edges in the triangle is of color α (otherwise together with u_1, we get a monochromatic triangle). Without loss of generality, assume u_2u_3 is a center edge of the triangle vu_2u_3v. Then $P_1 = vw_1w_2 \cdots w_t$ and the triangle vu_2u_3v form a nice shovel larger than Y(see Figure 2(b)), a contradiction. □



(a) $col(u_3u_1) \neq \alpha$ (b) $col(u_3u_1) = \alpha$

Figure 2: Cases for a larger nice shovel

Lemma 4. *Let G be a mono-C_3-free edge-colored K_n. Let S and Y be vertex-disjoint subgraphs of G such that $S = u_1u_2u_3u_1$ is a triangle with u_2u_3 being a center edge and Y is a nice shovel or a vertex set of size at most 2. Then G contains a PC path $P = z_1z_2\cdots z_\ell$ such that $V(P) = V(S) \cup V(Y)$, $z_1 = u_1$ and $col(z_1z_2) \in col(u_1, \{u_2, u_3\})$.*

Proof. We leave the easy case $|Y| \leq 2$ to be checked by readers. Now assume that Y is a nice shovel. Let $S' = v_1v_2v_3v_1$ be the unique triangle in Y with v_2v_3 being the center edge. Let $w_1 = v_1$ and $w_1w_2\cdots w_t$ be the unique Hamilton path in $Y - \{v_2, v_3\}$. Let $col(u_2u_3) = c_1$ and $col(v_2v_3) = c_2$. If there is an edge between $\{u_2, u_3\}$ and $\{v_2, v_3\}$ assigned a color distinct to both c_1 and c_2, then a desired PC path can be found immediately. This argument, together with the mono-C_3-free condition, implies that $col(u_2u_3) \neq col(v_2v_3)$. Without loss of generality, assume $col(u_2u_3) = 1$, $col(v_2v_3) = 2$ and $col(u_2v_2) = 1$. The mono-C_3-free condition again forces $col(u_3v_2) = 2$, $col(u_3v_3) = 1$ and $col(u_2v_3) = 2$ (see Figure 3).

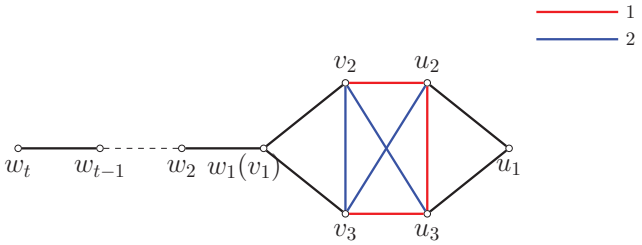


Figure 3: Colors between S and Y

If $col(u_1u_2) \neq 2$ or $col(v_1v_3) \neq 1$, then $u_1u_2v_3u_3v_2v_1w_2\cdots w_t$ or $u_1u_2v_2u_3v_3v_1w_2\cdots w_t$ is a desired PC path. For the remaining case that $col(u_1u_2) = 2$ and $col(v_1v_3) = 1$, we consider the PC path $P_1 = v_3v_1w_2\cdots w_t$. Since $col(u_3v_3) = col(v_3v_1) = 1$, by Lemma 1, G contains a PC path $P_1' = x_1x_2\cdots x_{t+2}$, where $V(P_1') = V(P_1)\cup\{u_3\}$, $x_1 = v_3$ and $col(x_1x_2) = col(v_3v_1)$. Then $u_1u_2v_2x_1x_2\cdots x_{t+2}$ is a desired PC path. This complets the proof. □

3 Directed almost spanning trees in mono-C_3-free tournament

Let G be an edge-colored graph. If there is a function $h : V(G) \to col(G)$ such that $col(uv) = h(u)$ or $h(v)$ for each edge $uv \in E(G)$. Then we can define a directed graph D_G with $V(D_G) = V(G)$ and $uv \in A(D_G)$ if and only if $col(uv) = h(u)$ and $col(uv) \neq h(v)$. Such an edge-colored graph G is called "degenerate" in [24]. An easy observation in [23] states that each PC cycle in G must be a directed cycle in D_G and vise versa. This one-to-one relationship does not hold if we study paths or trees instead of cycles. One can easily check that a PC path in G is not necessarily a directed path in D_G. However, certain oriented trees in D_G correspond to PC trees in G.

Let G be a mono-C_3-free edge-colored K_n. When G is degenerate, we will see that D_G acts like a tournament. In the following, we define such a digraph and obtain an oriented-tree-embedding result by several lemmas. This embedding result is applied in the proof of Theorem 2.

Definition 4. Given $t \geq 2$, let D be a multipartite tournament such that each partite set has at most 2 vertices and $N^+(x) \cap N^+(y) = \emptyset$ for every pair of distinct vertices x and y from a same partite set. Then we say D is a *mono-C_3-free tournament*.

Lemma 5 (Li [22])**.** *Given $t \geq 2$, let D be a strongly connected mono-C_3-free tournament with $|V(D)| \geq 4$, then each vertex of D is contained in directed cycles of lengths from 4 to $|V(D)|$.*

Lemma 6. *Given $t \geq 2$, let D be a mono-C_3-free tournament. Let $D_0, D_1, \ldots, D_p$ ($p \geq 0$) be strongly connected components of D such that for $0 \leq i < j \leq p$, either $D_j \to D_i$ or there is no arc between D_i and D_j. If $|V(D)| \geq 3$, then the following statements hold.*
(a) If $p = 0$, then D contains a directed Hamilton cycle.
(b) If $p \geq 1$ and $|N_D^+(v)| \geq 1$ for all $v \in V(D)$, then each partite set of size 2 is contained in D_0,

$D - D_0$ is a tournament and $D_j \to D_i$ for $0 \le i < j \le p$.
(c) If $|N_D^+(v)| \ge 1$ for all $v \in V(D)$, then D contains a directed Hamilton path.
(d) If there are at least two partite sets of size 2 in D, then all the partite sets of size 2 are contained in D_0, $D - D_0$ is a tournament and $D_j \to D_i$ for $0 \le i < j \le p$.

Proof. The case $|V(D)| = 3$ can be verified immediately. Assume $|V(D)| \ge 4$. Then the statement (a) holds by Lemma 5. To prove (b), suppose that some partite set $W = \{x, y\}$ is not contained in D_0. If $W \cap V(D_0) = \emptyset$, then each $z \in V(D_0)$ must be a common out-neighbour of x and y, a contradiction. If $W \cap V(D_0) \neq \emptyset$, say $W \cap V(D_0) = \{x\}$ and $y \in D_i$ for some $i \in [1, p]$, then by the condition that $|N_D^+(x)| \ge 1$, there must be a vertex $z \in V(D_0)$ such that $x \to z$. Note that D is a multipartite tournament, we have $y \to z$, a contradiction. Therefore each partite set of size 2 is contained in D_0, which implies that $D - D_0$ is a tournament and $D_j \to D_i$ for $0 \le i < j \le p$. So the statement (b) holds. The statement (c) can be obtained directly by statements (a) and (b). To prove (d), assume that $W_1, W_2, \ldots, W_s$ are partite sets of size 2 in D with $s \ge 2$ and $W_i = \{x_i, y_i\}$ $(i \in [1, s])$. For distinct indices a and b in $[1, s]$, let $D' = D[W_a \cup W_b]$. If there is a vertex $v \in V(D')$ such that $d_{D'}^-(v) = 2$, then either $v \in N^+(x_a) \cap N^+(y_a)$ or $v \in N^+(x_b) \cap N^+(y_b)$, a contradiction. So D' must be a directed cycle of length 4. By the arbitrariness of a and b, we know that $D[\cup_{i=1}^s W_i]$ is a strongly connected digraph. Therefore $D[\cup_{i=1}^s W_i] \subseteq D_0$ (otherwise each vertex $v \in V(D_0)$ is in $N^+(x_i) \cap N^+(y_i)$ for every W_i), which implies that $D - D_0$ is a tournament and $D_j \to D_i$ for $0 \le i < j \le p$. So the statement (d) holds. □

Lemma 7. *Let D be a mono-C_3-free tournament. Let $P = v_1 v_2 \cdots v_t$ be a directed path in D. Assume there exists a vertex $v \in V(D) \setminus V(P)$ such that $v \to v_i$ for some $i \in [1, t]$. Then D contains a directed path $P' = u_1 u_2 \cdots u_{t+1}$, where $V(P') = V(P) \cup \{v\}$ and $u_{t+1} = v_t$.*

Proof. Let j be a minimum integer in $[1, t]$ such that $v \to v_j$. If $j = 1$, then $P' = v v_1 \cdots v_{t-1} v_t$ is a desired directed path. If $j \ge 2$, then $v_j \in N^+(v) \cap N^+(v_{j-1})$. This implies that v and v_{j-1} belong to distinct partite set. By the assumption of j, we have $v v_{j-1} \notin A(D)$. Therefore $v_{j-1} \to v$ and $P' = v_1 \cdots v_{j-1} v v_j v_{j+1} \cdots v_t$ is a directed path in D. □

Given positive integers $\{l_i\}_{i=2}^k (k \ge 2)$, let P_i be a directed path of length l_i from u_i to u for each $i \in [2, k]$, satisfying $V(P_i) \cap V(P_j) = \{u\}$ for $2 \le i < j \le k$. Let x, y be two vertices not in $\cup_{i=2}^k V(P_i)$. Define a directed tree $T^*_{l_2,\ldots,l_k}$ (see Figure 4) as

$$V(T^*_{l_2,\ldots,l_k}) = \{x, y\} \cup \bigcup_{i=2}^{k} V(P_i) \quad \text{and} \quad A(T^*_{l_2,\ldots,l_k}) = \{ux, uy\} \cup \bigcup_{i=2}^{k} A(P_i).$$

We call u the *root* of $T^*_{l_2,\ldots,l_k}$.

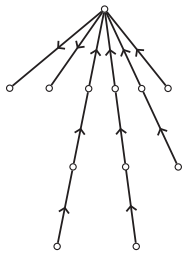


Figure 4: The tree $T^*_{3,3,2,1}$

Theorem 3. *For $k \ge 2$ and $\ell_2 \ge \ell_3 \ge \cdots \ge \ell_k \ge 1$, let D be a mono-C_3-free tournament on $n \ge \sum_{i=2}^k \ell_i + 2k^2 + 2k + 6$ vertices. If $|N_D^+(v)| \ge 2$ for all $v \in V(D)$. Then D contains a subgraph isomorphic to $T^*_{\ell_2,\ldots,\ell_k}$.*

Proof. We use $D_0, D_1 \ldots, D_p$ to denote the strongly connected components of D. Let $U = V(D) \setminus V(D_0)$. If $|U| \ge \sum_{i=2}^k \ell_i$, then $p \ge 1$ and by Lemma 6(b), we can assume that $D_j \to D_i$ for $0 \le i < j \le$

p and $D[U]$ is a tournament with P being a directed Hamilton path in $D[U]$. Decompose P into $k-1$ vertices disjoint directed paths $P_2, P_3, \ldots, P_k$ ending at $v_2, v_3, \ldots, v_k$ of length $\ell_2-1, \ell_3-1, \ldots, \ell_k-1$, respectively. Take a vertex $v_0 \in D_0$. Then $v_i \to v_0$ for all $2 \le i \le k$. Since $|N^+(v_0)| > 2$, there are two distinct vertices $x, y \in D_0$ satisfying $v_0 \to x$ and $v_0 \to y$. Those paths and together with x, y, v_0 form a tree $T^*_{\ell_2,\ldots,\ell_k}$. The remaining case is $|U| < \sum_{i=2}^k \ell_i$. Take $k-1$ integers $\ell'_2, \ldots, \ell'_k$ such that $0 \le \ell'_i \le \ell_i$ for each $i \in [2,k]$ and $\sum_{i=2}^k \ell'_i + |U| = \sum_{i=2}^k \ell_i$. Then $|V(D_0)| \ge \sum_{i=2}^k \ell'_i + 2k^2 + 2k + 6$.

Claim 1. If D_0 contains a subgraph T_1 isomorphic to $T^*_{\ell'_2,\ldots,\ell'_k}$, then D contains a subgraph isomorphic to $T^*_{\ell_2,\ldots,\ell_k}$.

Proof. Denote by u the root of T_1 and by $u_2, \ldots, u_k$ the k leafs of T_1 with $d^-_{T_1}(u_i) = 0$ for $2 \le i \le k$. Define $r_i = \ell_i - \ell'_i$ for $i \in [2,k]$. Since $|U| = \sum_{i=2}^k r_i$ and $D[U]$ is a tournament (by Lemma 6(b)), U can be partitioned into $k-1$ parts such that the i-th part is a tournament of order r_i. Note that for every pair of vertices $u \in U$ and $v \in D_0$, we have $u \to v$. Hence T_1 can be extended into a tree $T^*_{\ell_2,\ldots,\ell_k}$ by vertices in U. □

By Claim 1, it remains to prove that D_0 contains a $T^*_{\ell'_2,\ldots,\ell'_k}$ as a subgraph. Without loss of generality, assume that $\ell'_i \ge 1$ for $i \in [2,t]$. Let v be a vertex with minimum out-degree in D_0. Let $A = N^+_{D_0}(v)$ and $B = N^-_{D_0}(v)$. Then $|A| \le \frac{|D_0|-1}{2}$. Since each vertex in D_0 is adjacent to at least $|D_0|-2$ vertices, we have

$$|B| \ge |D_0| - 2 - |A| \ge |D_0| - 2 - \frac{|D_0|-1}{2} \ge \frac{|D_0|-3}{2} \ge k^2 + k + 1. \tag{1}$$

If $|B| \ge \sum_{i=2}^k \ell'_i$, then let P be a longest directed path in $D_0[B \cup \{v\}]$ with v being the ending vertex. By Lemma 7, P is a directed Hamilton path in $D_0[B \cup \{v\}]$. Decompose P into disjoint directed paths $P_2, \ldots, P_k$ ending at $v_2, v_3, \ldots, v_k$ of lengths $\ell'_2 - 1, \ell'_3 - 1, \ldots, \ell'_k - 1$, respectively. Recall that $|N^+_D(v)| \ge 2$ and $N^+_D(v) = N^+_{D_0}(v) = A$. There are two distinct vertices $x, y \in A = N^+(v)$. Note that $\{v_2, v_3, \ldots, v_k\} \subseteq N^-(v)$. We get a $T^*_{\ell'_2,\ldots,\ell'_k}$ as $(\cup_{i\in[2,k]}(P_i + v_iv)) \cup \{uw_1, uw_2\}$.

Now we assume $|B| \le \sum_{i=2}^k \ell'_i - 1$. This implies

$$|A| \ge |D_0| - 2 - |B| \ge 2k^2 + 2k + 4 + \sum_{i=2}^k \ell'_i - |B| \ge 2k^2 + 2k + 5. \tag{2}$$

Claim 2. If $D_0[A]$ contains $k-1$ disjoint directed paths $P_2, P_3, \ldots, P_k$ ending at vertices $a_2, a_3, \ldots a_k$ respectively and satisfying the following conditions:
(i) $1 \le |P_i| \le \max\{\ell'_i - 1, 1\}$ and $d^+_B(a_i) \ge k-1$ for $2 \le i \le k$;
(ii) there is a set $W \subseteq A \setminus \cup_{i=1}^k V(P_i)$ with $w \to z$ for every $w \in W$ and every $z \in \cup_{i=1}^k V(P_i)$ or $W = \emptyset$ such that

$$|W| + \sum_{i=2}^k |P_i| + |B| \ge \sum_{i=2}^k \max\{\ell'_i, 2\},$$

then D_0 contains a $T^*_{q_2,\ldots,q_k}$. Here $q_i = \max\{\ell'_i, 2\}$ for $2 \le i \le k$.

Proof. Condition (i) tells the existence of a $k-1$ matching $\{a_ib_i : 2 \le i \le k\}$ for a set $\{b_i : 2 \le i \le k\} \subseteq B$. Let u_i be the starting vertex of P_i and let $L_i = u_iP_ia_ib_iv$ for $2 \le i \le k$. Apparently L_i is a directed path of length between 2 and $\max\{\ell'_i, 2\}$. We assert that for each vertex $x \in (W \cup B) \setminus \{b_i : 2 \le i \le k\}$ and each L_i, there is a vertex $y^x_i \in V(L_i)$ such that $x \to y^x_i$. This can be done by setting $y^x_i = v$ when $x \in B \setminus \{b_i : 2 \le i \le k\}$ and setting $y^x_i = a_i$ when $x \in W$. Note that $|W| + \sum_{i=2}^k |P_i| + |B| \ge \sum_{i=2}^k \max\{\ell'_i, 2\}$. By Lemma 7, we can extend each path L_i first by vertices in $B \setminus \{b_i : 2 \le i \le k\}$ and then by vertices in W until each leg reach a length of $\max\{\ell'_i, 2\}$. Recall that $|A| \ge 2k^2 + 2k + 4 + \sum_{i=2}^k \ell'_i - |B|$. There are at least $2k^2$ vertices in A not added into legs. Choose two of them as out-neighbors of v. Then we find a $T^*_{q_2,\ldots,q_k}$. □

To satisfy the condition of Claim 2, we need to analyse the structure of $D_0[A]$. If $D_0[A]$ is a tournament or $D_0[A]$ has at least two partite sets of size two, then set $A' = A$. If there is exactly one partite set of size two contained in $D_0[A]$, say $W = \{x, y\}$, then set $A' = A - x$. In both cases, we can assume that $Q_1, Q_2, \ldots, Q_t (t \geq 1)$ are strongly connected components of $D_0[A']$ with $Q_j \to Q_i$ for $1 \leq i < j \leq t$ (by Lemma 6), $A' \subset A$ and $|A'| \geq |A| - 1 \geq 2k^2 + 2k + 4$. Now we analyse $D_0[A']$ instead of $D_0[A]$. The following claim tells that either some Q_j is large or there is an index j such that both $\sum_{i \leq j} |Q_i|$ and $\sum_{i > j} |Q_i|$ are large.

Claim 3. There exists some $j \in [1, t]$ such that one of the following holds
(i) $\sum_{i=1}^{j} |Q_i| \geq k - 1$ and $\sum_{i=j+1}^{t} |Q_i| \geq k - 1$;
(ii) $|Q_j| \geq |A'| - 2k + 4$;

Proof. Recall that $|A'| \geq 2k^2 + 2k + 4$. Let j be the smallest index in $[1, t]$ such that $\sum_{i=1}^{j} |Q_i| \geq k - 1$. So $\sum_{i=1}^{j-1} |Q_i| \leq k - 2$. If (i) does not hold, then $\sum_{i=j+1}^{t} |Q_i| \leq k - 2$. Therefore

$$|Q_j| \geq |A'| - \sum_{i=j+1}^{t} |Q_i| - \sum_{i=1}^{j-1} |Q_i| \geq |A'| - (k-2) - (k-2) = |A'| - 2k + 4.$$

□

Now we proceed the proof by analyzing the two cases in Claim 3.

Case 1. There exists some $j \in [1, t]$ such that $\sum_{i=1}^{j} |Q_i| \geq k - 1$ and $\sum_{i=j+1}^{t} |Q_i| \geq k - 1$

In this case, apply Lemma 6(c) to $D_0[\cup_{i=1}^{j} Q_i]$, we get a directed Hamilton path P in $D_0[\cup_{i=1}^{j} Q_i]$. Note that $N^+_{D_0}(z) \subseteq B \cup \left(\cup_{i=1}^{j} Q_i\right) = B \cup V(P)$ for each vertex $z \in V(P)$ and v is a vertex in D_0 with minimum out-degree. We have

$$d^+_{D_0}(z, B) \geq d^+_{D_0}(z) - |P| \geq d^+_{D_0}(v) - |P| = |A| - \sum_{i=1}^{j} |Q_i| \geq \sum_{i=j+1}^{t} |Q_i| \geq k - 1.$$

Set $W := \cup_{i=j+1}^{t} Q_i$. If $|P| \geq \sum_{i=2}^{k} \max\{\ell'_i - 1, 1\}$, then find $k - 1$ disjoint paths $P_2, P_2, \ldots, P_k$ in P such that $|P_i| = \max\{\ell'_i - 1, 1\}$ for $i \in [2, k]$. Then

$$|W| + \sum_{i=2}^{k} |P_i| + |B| > \sum_{i=2}^{k} |P_i| + k - 1 \geq \sum_{i=2}^{k} \max\{\ell'_i, 2\}.$$

If $|P| < \sum_{i=2}^{k} \max\{\ell'_i - 1, 1\}$, then let $P_2, P_3, \ldots, P_k$ be a decomposition of P into $k - 1$ disjoint paths such that $1 \leq |P_i| \leq \max\{\ell'_i - 1, 1\}$ for $i \in [2, k]$ (this is possible since $|P| = \sum_{i=1}^{j} |Q_i| \geq k - 1$). By (2), we have

$$|W| + \sum_{i=2}^{k} |P_i| + |B| = |A'| + |B| \geq \sum_{i=2}^{k} \ell'_i + 2k^2 + 2k + 3 > \sum_{i=2}^{k} \max\{\ell'_i, 2\}.$$

In both situations, we obtain a tree $T^*_{\ell'_2, \ldots, \ell'_k}$ by Claim 2.

Case 2. There exists some $j \in [1, t]$ such that $|Q_j| \geq |A'| - 2k + 4$.

In this case, by Lemma 5, Q_j contains a directed Hamilton cycle $C = w_1 w_2 \cdots w_h w_1$. Let $h_2, h_3, \ldots, h_k$ be positive integers such that $h_i \leq \max\{\ell'_i - 1, 1\}$ for $2 \leq i \leq k$, $\sum_{i=2}^{k} h_i \leq |C| = h$ and $\sum_{i=2}^{k} h_i$ is maximum. We say a sequence of indices $(\alpha_2, \alpha_3, \ldots, \alpha_k)$ is a *model* on C if $\alpha_{j+1} = \alpha_j + h_j$ for $2 \leq j \leq k - 1$. Here the addition is on $\mathbb{Z}_h$. Given a model, we can find vertex disjoint

paths $P_2, P_3, \ldots, P_k$ such that P_i is of length $h_i - 1$ and ending at the vertex w_{α_i}. Note that $\sum_{i=2}^{k} |P_i| \geq \min\left\{h, \sum_{i=2}^{k}(\ell'_i - 1)\right\}$. We have

$$\begin{aligned}\sum_{i=2}^{k} |P_i| + |B| &\geq \min\left\{h, \sum_{i=2}^{k}(\ell'_i - 1)\right\} + |B| \\ &\geq \min\left\{|A'| - 2k + 4, \sum_{i=2}^{k}(\ell'_i - 1)\right\} + |B| \\ &\geq \min\left\{|A| + |B| - 2k + 3, \sum_{i=2}^{k}(\ell'_i - 1) + |B|\right\}.\end{aligned}$$

By (1) and (2), there holds $\sum_{i=2}^{k} |P_i| + |B| \geq \sum_{i=2}^{k} \max\{\ell'_i, 2\}$. If there is a model $(\alpha_2, \alpha_3, \ldots, \alpha_k)$ such that $d^+_{D_0}(w_{\alpha_i}, B) \geq k - 1$ for $2 \leq i \leq k$. We can obtain a $T^*_{\ell'_2,\ldots,\ell'_k}$ by Claim 2.

Now assume that for every model $(\alpha_2, \alpha_3, \ldots, \alpha_k)$, there exists a vertex w_{α_i} such that $d^+_{D_0}(w_{\alpha_i}, B) \leq k - 2$. Let $M_0 = (\alpha_2, \alpha_3, \ldots, \alpha_k)$ be a model on C. Define $M_j = (\alpha_2 + j, \alpha_3 + j, \ldots, \alpha_k + j)$ for $1 \leq j \leq h - 1$. Here the addition is on $\mathbb{Z}_h$. Then M_j is also a model on C. Let $\mathcal{M} = \{M_i : 0 \leq i \leq h - 1\}$. Let

$$X = \{\alpha : d^+_{D_0}(w_\alpha, B) \leq k - 2\}.$$

Note that each $\alpha \in X$ lies in exactly $k - 1$ models in $\mathcal{M}$, and for each model M_i, there is at least one entry belonging to X. Therefore

$$|X| \geq \frac{|\mathcal{M}|}{k-1} = \frac{h}{k-1} \geq \frac{|A'| - 2k + 4}{k-1}.$$

Let H be the induced subgraph of D_0 on $\{w_\alpha : \alpha \in X\}$. Then there is a vertex z in H with $d^+_H(z) \leq \frac{|X|-1}{2}$. Then

$$\begin{aligned}|A| = d^+_{D_0}(v) \leq d^+_{D_0}(z) &\leq d^+_H(z) + |A \setminus X| + d^+_{D_0}(z, B) \\ &\leq \frac{|X| - 1}{2} + |A| - |X| + k - 2.\end{aligned}$$

Hence $|X| \leq 2k - 5$. Recall that $|X| \geq \frac{|A'|-2k+1}{k-1} \geq \frac{|A|-2k}{k-1}$. We get

$$|A| \leq 2k^2 - 5k + 5,$$

which contradicts (2) that $|A| \geq 2k^2 + 2k + 5$. The proof is complete. □

4 Proofs of Theorem 1 and 2

Recall that $g(S_k, C_3)$ is the maximum number N such that there exists an edge-colored K_N containing neither a rainbow S_k nor a monochromatic C_3. We will use this notation to state the proofs of main theorems.

Proposition 1. *$g(S_k, C_3) < (k + 1)!$ for every integer $k \geq 1$.*

Proof. We prove by induction on k. When $k = 1$, it is trivial. Assume that the statement holds for $g(S_{k-1}, C_3)$. Let G be a mono-C_3-free edge-colored K_n with $n \geq (k+1)!$. By the inductive hypothesis, G contains a rainbow $(k-1)$-star with v being the vertex of degree $k - 1$. If $d^c_G(v) \geq k$, then the proof is complete. If $d^c_G(v) \leq k - 1$, then there exists a vertex set W of size at least $\frac{(k+1)!-1}{k-1} \geq k!$. Since $|W| \geq k!$, by the inductive hypothesis and the assumption that G is mono-C_3-free, $G[W]$ contains a rainbow $(k-1)$-star T with u being the center vertex of degree $k - 1$. Note that G has no monochromatic triangle, we know that $col(v, W) \cap col(T) = \emptyset$. So $T + uv$ is a rainbow k-star. □

Lemma 8. *Given a positive integer $k \geq 1$, let G be a mono-C_3-free edge-colored K_n with $n \geq (k+2)!$, then G contains a PC copy of every tree of k edges.*

Proof. By induction on k. The case $k = 1$ can be verified immediately. Assuming that the conclusion holds for every tree of $k-1$ edges. Let T be a tree of k edges with a leaf vertex x and a pendent edge xy. Define $V_0 = \{v \in V(G) : d^c(v) \leq k-1\}$ and $V_1 = V(G) \setminus V_0$. Since $G[V_0]$ contains no rainbow S_k, by Proposition 1, we have $|V_0| < (k+1)!$. Therefore $|V_1| > (k+2)! - (k+1)! > (k+1)!$. By the inductive hypothesis, $G[V_1]$ contains a subgraph T' which is a PC copy of $T-x$. Let $\varphi : V(T) \setminus \{x\} \to V(T')$ be an isomorphism mapping from $T-x$ to T'. Let $y' = \varphi(y)$. By the definition of V_1, we have $d^c_G(y') \geq k$. Note that $|T'-y| \leq k-1$. There exists a vertex $x' \in V(G) \setminus V(T')$ such that $col(y'x') \notin col(y', V(T'))$. Hence $T' + y'x'$ is a PC copy of T in G. The proof is complete. □

Proof of Theorem 1. By Lemma 8, G contains a PC copy of T_0. Denote by T_1 a maximal PC subgraph of G such that T_1 is a subdivision of T_0. If $V(T_1) = V(G)$, then we are done. Otherwise, for each edge $xy \in E(T_1)$, denote by $T_1(x, y)$ and $T_1(y, x)$ the two connected components of $T_1 - xy$ containing x and y, respectively. For a vertex $u \in V(G) \setminus V(T_1)$, if $col(ux) = col(xy)$, then we say the triple (u, x, y) is *compatible* to T_1. By the assumption that T_1 is maximal, for every leaf vertex x and the pendent edge xy in T_1, the triple (u, x, y) is compatible. Now let (u, x, y) be a triple compatible to T_1 with $|T_1(y, x)|$ being the minimum (here x may not be a leaf vertex). Since G is mono-C_3-free and $col(ux) = col(xy)$, we have $col(uy) \neq col(ux)$. If $col(uy) \notin col(y, N_{T_1(y,x)}(y))$, then $T_1 - xy + xu + uy$ forms a PC subdivision of T_0 larger than T_1, a contradiction. Hence there exists a vertex $z \in N_{T_1(y,x)}(y)$ with $col(uy) = col(yz)$. This implies that (u, y, z) is compatible to T_1 with $|T_1(z, y)| < |T_1(y, x)|$, a contradiction. □

Proof of Theorem 2. The proof is given by distinguishing the number of vertex-disjoint nice bowties in G.

Case 1. There are at least $g(S_k, C_3) + 1$ vertex-disjoint nice bowties in G.

Let $B_1, \cdots, B_s$ be vertex-disjoint nice bowties in G with $s \geq g(S_k, C_3)+1$ such that $G - \cup_{i=1}^{h} V(B_i)$ contains no nice bowtie. Pick one center v_i from each bowtie B_i $(1 \leq i \leq s)$ to form a set W. Since G is mono-C_3-free and $|W| \geq g(S_k, C_3) + 1$, there exists a rainbow k-star S_k in $G[W]$. Without loss of generality, assume $V(S_k) = \{v_0, v_1, v_2, \ldots, v_k\}$ and v_0 is the k-degree vertex in S_k.

For $1 \leq i \leq k$, since v_i is a center vertex of the bowtie B_i, there exists a subgraph $X_i \subseteq B_i$ such that $X_i + v_iv_0$ is either a nice 2-shovel or a nice 3-shovel. Denote by Y_i the nice shovel $X_i + v_iv_0$. Then $Y_1, Y_2, \ldots, Y_k$ are disjoint shovels overlapping at the vertex v_0. We will obtain a desired PC spider by extending or shrinking these shovels.

Define $I = \{1, 2, \ldots, k\}$ and $I^- = \{i \in I : \ell_i \leq |Y_i| - 1\}$. For each $i \in I^-$, we can modify Y_i into a leg of length ℓ_i by removing $|Y_i| - 1 - \ell_i$ vertices. Let R be the removed vertices. Then $U = (V(G) \setminus \cup_{i=1}^{k} V(Y_i)) \cup R$ is the set of vertices waiting to be embedded. Note that $|U| = \sum_{i \in I \setminus I^-} (\ell_i - |Y_i| + 1)$. We partition U into $|I \setminus I^-|$ disjoint sets $\{U_i\}_{i \in I \setminus I^-}$ such that $|U_i| = \ell_i - |Y_i| + 1$ for each $i \in I \setminus I^-$. Then by Lemma 3, each $G[U_i]$ of order at least 3 contains a spanning nice shovel F_i. Set $F_i = U_i$ when $|U_i| \leq 2$. Apply Lemma 4 to the triangle in Y_i and F_i for all $i \in I \setminus I^-$, we get a desired PC spanning tree in G.

Case 2. The number of vertex-disjoint nice bowties in G is at most $g(S_k, C_3)$.

In this case we will first find a structure that is almost the spider we want except the first leg, which is a nice 1-shovel. Then we get the final spider by extending this shovel into a leg of length ℓ_1. To state the proof, we need more definitions. Let S be a spider with precisely $k-1$ legs of lengths $\ell_2, \ell_3, \ldots, \ell_k$ respectively, let Y be a triangle and let $O^*_{\ell_2,\ldots,\ell_k}$ be the graph obtained by identifying a vertex in Y with the center of S. We call $O^*_{\ell_2,\ldots,\ell_k}$ an *octopus*. The legs of $O^*_{\ell_2,\ldots,\ell_k}$ are exactly the legs of S. We say $O^*_{\ell_2,\ldots,\ell_k}$ is *nice* if S is a PC spider and $Y \cup P$ is a nice shovel for each leg P of S. If G contains a nice octopus $O^*_{\ell_2,\ldots,\ell_k}$, then let $F = G - O^*_{\ell_2,\ldots,\ell_k}$. By Lemma 3, F contains a spanning nice shovel F' when $|F| \geq 3$. Set $F' = F$ when $|F| \leq 2$. Apply Lemma 4 to F' and the triangle in $O^*_{\ell_2,\ldots,\ell_k}$, we get a desired PC spider. Therefore the following claim holds.

Claim A. If G contains a nice octopus $O^*_{\ell_2,\ldots,\ell_k}$, then G contains a PC spider with legs of lengths $\ell_1, \ell_2, \ldots, \ell_k$.

Let $B_1, B_2, \ldots, B_s$ be vertex-disjoint nice bowties in G such that there is no nice bowtie in $H := G \setminus \cup_{i=1}^{s} V(B_i)$. Then $s \leq g(S_k, C_3)$. Recall that each bowtie has at most 6 vertices. We have $|H| \geq n - 6g(S_k, C_3)$. Note that $\ell_1 \geq \frac{n-1}{k} \geq 6g(S_k, C_3) + 2k^2 + 2k + 7$. Then

$$|H| = 1 + \sum_{i=2}^{k} \ell_i + \ell_1 - 6g(S_k, C_3) \geq \sum_{i=2}^{k} \ell_i + 2k^2 + 2k + 8.$$

For each vertex v in H, we claim that at most one color in $col(v, N_H(v))$ appears more than once at v. Otherwise by the mono-C_3-free condition, H contains a nice short bowtie with v being the center vertex, which contradicts the choice of H.

Claim B. For each vertex $v \in V(H)$, at most one color in $col(v, N_H(v))$ appears more than once at v.

There could be a vertex v with no color repeats at v. Let $V_1 = \{v \in V(H) : \Delta_H^{mon}(v) = 1\}$ and $V_2 = \{v \in V(H) : \Delta_H^{mon}(v) \geq 2\}$.

If $|V_1| \geq 3$, then let v_1, v_2, v_3 be distinct vertices in V_1. By the definition of V_1, the cycle $v_1v_2v_3v_1$ must be a rainbow triangle. Let $H' = H - \{v_1, v_2, v_3\}$. Recall that $|H| \geq \sum_{i=2}^{k} \ell_i + 2k^2 + 2k + 8$. We have $|H'| \geq \sum_{i=2}^{k} \ell_i + 2k^2 + 2k + 5$. Let $X_2, X_3, \ldots, X_k$ be disjoint sets in H' such that $|X_i| = \ell_i$ for $2 \leq i \leq k$. By Lemma 2, there is a PC path P_i on $\{v_1\} \cup X_i$ with v_1 being the starting vertex for $2 \leq i \leq k$. Since $\Delta_{H'}^{mon}(v_1) = 1$, the triangle $v_1v_2v_3v_1$ and paths $P_2, P_3, \ldots, P_k$ form a nice octopus $O^*_{\ell_2,\ldots,\ell_k}$. By Claim A, we are home.

If $2 \leq \Delta_H^{mon}(x) \leq 2k^2 + 2k + 7$ for some vertex $x \in V_2$, then assume α is the color such that $\Delta_H^{mon}(x) = |\{u \in V(H) : col(ux) = \alpha\}|$. Let $U_\alpha = \{u \in V(H) : col(ux) = \alpha\}$ and $H' = H - U_\alpha$. Then $|H'| \geq \sum_{i=2}^{k} \ell_i + 1$. Let $X_2, X_3, \ldots, X_k$ be disjoint sets in $H' - x$ such that $|X_i| = \ell_i$ for $2 \leq i \leq k$. By Lemma 2, there is a PC path P_i on $\{x\} \cup X_i$ with v_1 being the starting vertices for $2 \leq i \leq k$. Note that $\Delta_{H'}^{mon}(x) = 1$. Choose distinct vertices $a, b \in U_\alpha$, then the triangle $xabx$ and paths $P_2, P_3, \ldots, P_k$ form a nice octopus $O^*_{\ell_2,\ldots,\ell_k}$. By Claim A, we are home.

The remaining case is that $|V_1| \leq 2$ and $\Delta_H^{mon}(x) \geq 2k^2+2k+8$ for every $x \in V_2$. Let $H_1 = H - V_1$. Then $|H_1| \geq \sum_{i=2}^{k} \ell_i + 2k^2 + 2k + 6$ and $\Delta_{H_1}^{mon}(u) \geq 2k^2 + 2k + 6$ for every vertex $u \in V(H_1)$. Let $f(u)$ be the unique color appearing more than once in $col(u, N_{H_1}(u))$. We claim that for any two vertices $u, v \in V(H_1)$, there holds $col(u) = f(u)$ or $f(v)$. Otherwise, H contains a nice long bowtie with u and v being the center vertices, which contradicts the choice of H.

Claim C. For each pair of vertices $u, v \in V(H)$, we have $col(uv) = f(u)$ or $f(v)$.

Now construct an auxiliary digraph D satisfying $V(D) = V(H_1)$ and $A(D) = \{uv : col(uv) = f(u)$ and $col(uv) \neq f(v)\}$. Since H_1 is mono-C_3-free, according to Claim C and Definition 4, the directed graph D is a mono-C_3-free tournament with $N_D^+(u) \geq 2k^2 + 2k + 6 > 2$ for every vertex $u \in V(H_1)$. Recall that $|H_1| \geq \sum_{i=2}^{k} \ell_i + 2k^2 + 2k + 6$. By Theorem 3, D contains a subgraph $T^*_{\ell_2,\cdots,\ell_k}$, which is a nice octopus $O^*_{\ell_2,\ldots,\ell_k}$ in G. Again we obtain the desired PC spanning tree by Claim A. The proof is complete. □

5 Conclusion

Tree embedding has become an increasingly popular research area in recent years. It has been extensively studied in various contexts, such as spanning trees or almost spanning trees in random graphs [26], random directed graphs [27], (n, d, λ) graphs [18, 19], tournaments [6, 7], dense directed graphs [21], edge-colored graphs [15] and finite vector spaces[10]. This paper demonstrates that in edge-colored complete graphs, the absence of monochromatic triangle implies the existence of PC copies of every spanning tree, which is a subdivision of a k-star. The most challenging part in the proof is the case of multipartite tournaments. In fact, every n-vertex rooted tree oriented from each child-vertex to its father-vertex, is contained in a transitive tournament D on n vertices. Color each arc in D with the label of its tail, resulting an edge-colored complete graph, which is mono-C_3-free and contains PC copies of every spanning tree. Motivated by this example, we ask the following question.

Question 1. Let T be an n-vetrex tree with maximum degree at most Δ and let G be a mono-C_3-free edge-colored K_n. Does G always contain a PC copy of T when n is sufficiently large?

A graph G is called $\mathcal{T}(n, \Delta)$-universal if G contains every spanning tree with maximum degree at most Δ. The statement of above question is also inspired by a question from Alon, Krivelevich and Sudakov[2] on the $\mathcal{T}(n, \Delta)$-universal property of (n, d, λ)-graphs. We also observed that the result of Benford and Montgomery[6] may help dealing with the multipartite tournament case in further research on Conjecture 1. It would be interesting to see more connections between Conjecture 1 and tree embedding results in graphs and digraphs.

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